PLANE GEOMETRY

1. **Geometry** : It is the science which treats of the shape , size and position of figures by pure reasoning of the definitions , axioms and established geometrical facts.
2. Volume : The volume of a body is the amount of space it occupies.
3. A **point** has position only. It has no length , no breadth and no thickness.
4. A **line** has length but is said to have no breadth.
5. Line **segment:** The line segment AB is defined as a limited portion of the straight line.
6. Length **of a line segment :** The length of a line segment is defined as the distance between its end points.
7. A **surface** has length and breadth but no thickness.
8. A **straight line** has the same direction through out its whole length.
9. A **curved line** changes its direction continuously from point to point.
10. **Plane surface** is a surface which is such that the straight line joining any two points in it lies wholly on the surface.
11. **Curved surface:** If straight line joining the points is not at all on the surface then such surface is called curved surface.
12. **Axiom** : All mathematical reasoning is based on certain basic assumptions .Such assumptions are called axioms.

    (or)Axiom is a principle which is universally accepted without proof.
13. **Hypothesis:** Hypothesis is the given information in the proposition.
14. **Proof :** It is the part of proposition in which truth of a theorem is established.
15. **Theorem**: A theorem is a proposition which some geometrical property is to be proved.
16. **Corollary**: A corollary is a statement , the truth of which readily follows from an established proposition.
17. **Angle**: **A** plane figure consisting of two distinct rays with a common origin and a plane region bounded by them is called an angle.
18. **Vertex:** Two straight lines are called the **arms** of the angle and the point which they meet is the **vertex** of the angle.

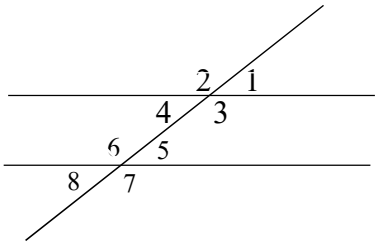
19. **Acute angle(<90^0) ,Right angle(90^0) ,Obtuse angle (90^0 < A < 180°)**

20. **Supplamentary angles** :If sum of two angles is equal to 180^0 then angles are supplementary.

21. **Complementary angles:** If sum of two angles is 90^0 then angles are complementary.

22. **Parallel lines: Two** lines are parallel if
(i)they lie in the same plane and
(ii)they do not intersect how ever far when they extended.

23. A line which intersects two given lines in distinct points is called **transversal** of those lines.



24. 1, 2, 7, 8 are **exterior angles.**
3, 4, 5, 6 are **interior angles.**
4, 5 ; 3, 6 ..are **alternate angles.**
1, 5 ; 2 ,6 ..are **corresponding angles.**

25. AB is straight line. If the point C is taken in between the points A and B then 'C' divides AB **internally** in the ratio **AC :CB**.

26. AB is a straight line.If a point C is taken on AB produced then AB is said to be divided externally at 'C' in the ratio **AC:BC.**

27. **Triangle** : A plane figure bounded by three straight lines is triangle.

28. **Perimeter of the triangle: T**he perimeter of a triangle is the sum of its sides.

29. **Exterior angle ,interior adjacent angle and interior opposite angles:**When one side of a triangle is produced, the angle formed out side the triangle is called the exterior angle. The angle adjacent to it is called the interior adjacent angle and the other two angles of the triangle are called interior opposite angles.
1. exterior angle , 2 . interior adjacent angle.
3 , 4 are Interior opposite angles.

30. **Altitiude**.The perpendicular drawn from any vertex upon the opposite side is called altitude.

31. **Median** : The line joining a vertex to the middle point of the opposite side is called median.

32. **Angle bisector** :**A** straight line or a plane that divides an angle into two equal parts.
33. **Similar triangles: Two** triangles are said to be similar if
    i) their corresponding angles are equal.
    ii) their corresponding sides are proportional.
34. **Equilateral triangle:** An equilateral triangle is one which has three sides of it are equal.
35. **Isosceles triangle:**An Isosceles triangle is one which has its two sides are equal.
36. **Scalene triangle** : A scalene triangle is one which has three unequal sides.
37. **Right angled triangle** : If one of the angle of triangle is 90^0 then it is right angled triangle.
38. **Obtuse angled triangle** : If one of the angle of triangle is obtuse then it is obtuse angled triangle.
39. **Acute angled triangle** : If all the three angles of triangle are acute then the triangle is acute angled triangle.
40. **Quadrilateral:** A plane figure bounded by four straight lines is called a quadrilateral.
41. **Parallelogram**: A quadrilateral having both pairs of opposite sides parallel is called parallelogram.
42. **Rectangle**:A parallelogram whose angles are right angles is called rectangle.
43. **Square**: A square is a rectangle whose sides of it are equal.
44. **Rhombus**: Rhombus is a quadrilateral which has all its sides are equal but its angles are not right angles.
45. **kite : A** kite is a quadrilateral formed by two isosceles triangle standing on the opposite sides of a common base.
46. **Cyclic quadrilateral:** A quadrilateral is called cyclic if all the four of its vertices lie on the circle.
47. **Circle:A Circle** is plane figure bounded by one curved line and is such that all straight lines drawn to this line from certain point with in it are equal.

48. **Circumference:** The **circumference** of a circle is the curved line bounding the circle.
49. The **arc** of a circle is any part of the circumference.
50. **A radius** of the circle is a straight line drawn from the centre to the circumference.
51. **A diameter** of a circle is a straight line drawn through the centre of a circle and terminated by the circumference.
52. **A semicircle** is a figure bounded by a diameter of a circle and part of the circumference cutoff by the diameter.
53. A **Chord** of a circle is a straight line joining any two points on its circumference.
54. A chord of a circle which does not passes through the centre divides the circle into two unequal arcs; of these , the greater is called the **major arc** and the lesser the **minor arc.**
55. **A Sector** of a circle is the area bounded between two radii and the arc intercepted between them.
56. **Sector angle:**The angle subtended by the arc of a sector at the centre is called the sector angle.
57. **A Segment** of a circle is the area bounded by the arc and its chord.
58. A **secant** of a circle is a straight line of indefinite length which cuts the circumference of the circle at two points.
59. A **tangent** to a circle is a straight line which meets the circumference of the circle at one and only one point.
60. **Common tangent: A** single line that forms a tangent to two or more seperate curves.
61. **Direct common tangent: I**f the points of contact lie on the same side of the line of centres , then such tangent is known as the **direct common tangent.**
62. **Transverse common tangent: If** the points of contact lie on opposite sides of the line of centres **,** then such tangent is known as **transverse common tangent.**
63. Circumcircle: A circle passing through all the vertices of a polygon is called a circumcircle.

64. **Incircle:** A circle touching all the sides of a polygon is called an incentre.

65. **Concentric circles:**Circles which have the same centres are called concentric circles.

66. **Alternate segment:** The segment opposite to the angle formed by the chord of a circle with the tangent at a point is called the alternate segment of the circle.

$\angle BSA$, $\angle BAT$ are angles in the alternate segment.

67. A **polygon** is a plane figure bounded by three or more straight lines.

68. A **polygon** is said to be equilateral when its sides are equal.

69. A **polygon** is said to be equiangular when all its angles are equal.

70. A **polygon** is regular when all its sides and all its angles are equal.

71. A **polygon** is said to be **convex** when each of its angles is less than two right angles.

72. A **polygon** is said to be **concave** when one or more of its angles are greater than two right angles.

Note *: A polygon unless otherwise mentioned is usually taken to mean a convex polygon.*

73. The join of any two vertices of polygon which are not consecutive is a **diagonal** of the polygon.

74. *Sum of all the interior angles of a convex polygon of 'n' sides is (2n - 4)right angles.*

75. Sum of the exterior angles of convex polygon is 360°.

76. **Inscribed polygon**: **A** polygon whose all vertices belong to a circle is said to be inscribed into this circle, and the circle is said to be circumscribed about this polygon.

77. **Circumscribed polygon**: A polygon whose all sides are tangent to a circle is said to be circumscribed about this circle , and the circle is said to be inscribed in this polygon.

78. **Axis of symmetry** : **A** figure is said to be symmetrical about a line called the (axis of symmetry) if the two parts of the figure on each side of it exactly coincide when folded about that line.

79. **Isosceles trapezium**: A trapezium whose nonparallel sides are equal known as isosceles trapezium.

80. **Cevian : A** cevian is a segment which joins the vertex of a triangle with a points on its opposite side(or its extension).

FORMULAE

1. Area of a triangle = (1/2)x base x altitude.

2. Area of an equilateral triangle = $\frac{\sqrt{3}}{2}a^2$ or $\frac{h^2}{3}$. Where 'a' is length of the base and ' h' is altitude.

3. Area of a scalene triangle = $\sqrt{s(s-a)(s-b)(s-c)}$ where $s = \frac{a+b+c}{2}$ where 'a' , 'b' and 'c' are sides of a triangle.

4. Length of median from vertex 'C' of triangle ABC is $\frac{\sqrt{2a^2+2b^2-c^2}}{2}$

5. Length of altitude from the vertex 'C' of triangle ABC is $\frac{\sqrt{(a+b-c)(b+c-a)(c+a-b)(a+b+c)}}{2c}$

6. Area of parallelogram = Base x height

7. Perimeter of a rectangle = 2(*1* + b) , Area of rectangle = *1* x b.

8. Perimeter of a square = 4 x side , Area of a square = side x side

9. Perimeter of a Rhombus = 4 x side.

10. Area of Rhombus = $\frac{d_1 d_2}{2}$. Where d_1 and d_2 are the diagonals.

11. Area of a trapezium = $\frac{h}{2}(a+b)$. ' a ' , ' b ' are lengths of parallel sides.

12. Area of a quadrilateral = $\frac{1}{2}(h_1+h_2)d$ .Where **d** is the diagonal and h_1, h_2 are the perpendiculars drawn to the diagonal.

13. Area of circle = πr^2 where 'r' is the radius of the circle.

14. Area of sector OAB = $\frac{lr}{2}$ where l = length of the arc, r = radius of

the circle.= $\frac{\angle AOB}{360^\circ}$ x Area of the circle.

= $\frac{1}{2}r^2\theta$ where θ is sector angle.

15. Circumference of a circle = $2\pi r$ where ' r ' is radius of the circle.

16. Length of direct common tangent = $\sqrt{d^2-(R-r)^2}$

17. Length of transverse common tangent = $\sqrt{d^2-(R+r)^2}$

(where 'R' , 'r' are radii of different circles 'd' is distance between the circles.)

18. A , B are centres of two circles, r_1 , r_2 are radii of two different circles. AB = Distance between the centers.

D.C = Direct common tnagent T.C = Transverse common tangent

S.No	Nature of circles	Condition	Number of D.C	Numer of T.C
(i)	Externally touch each other	AB = $r_1 + r_2$	2	1
(ii)	Internally touch each other.	AB = $\lvert r_1 - r_2 \rvert$	1	0
(iii)	Intersecting	AB < $r_1 + r_2$	2	0
(iv)	Non-intersecting	AB > $r_1 + r_2$	2	2

19. Cube.

i) lateral surface area = $4x^2$ (where ' x ' is side of the circle)

ii) total surface area = $6x^2$

iii) volume = x^3 where 'x' is the side of a cube.

20. **Cuboid**

i) Lateral surface area = $2(lh + bh)$

ii) Total surface area = $2(lh + bh + lb)$

iii) Volume = lbh (where l , b , h are length , breadth and height of the circle)

21. **Cone** (r = radius of the base, l = *slant height , h = height of the cone)*

i) Lateral surface area = πrl

ii) Total surface area = $\pi rl + \pi r^2$

iii) Volume = $\frac{1}{3}\pi r^2 h$

22. **Cylinder** (h = height of the cylinder, r = Radius of the base of cylinder)

i) Lateral surface area = $2\pi rh$

ii)Total surface area = $2\pi rh + 2\pi r^2$

iii) volume = $\pi r^2 h$

23. **Sphere** (' r ' is the radius of the sphere)

i)Lateral surface area = $4\pi r^2$

ii)Total surface area = $4\pi r^2$

iii) volume = $\frac{4}{3}\pi r^3$

24. In triangle ABC , AB = c , BC = a , CA = b are sides of triangle then

(i) Sine rule = $\frac{a}{SinA} = \frac{b}{SinB} = \frac{c}{SinC}$ = 2R.(Where R is circum radius)

(ii) Cosine rule $a^2 = b^2 + c^2 - 2bc\ CosA$,.....

25. If ' r ' is radius of incircle of the triangle ABC and Δ is area of the triangle then Δ = rs .

26. $\Delta = \frac{bcSinA}{2} = \frac{acSinB}{2} = \frac{baSinC}{2}$

Theorem statements

1. For any three points A , B , C the distance AB is not less than the difference of the distances AC and BC.

2. If transversal cuts the two parallel lines , then two

(i) corresponding angles are equal.

(ii) Alternate angles are equal.

(iii) The interior angles on the same side of the transversal are together equal to two right angles.

3. Any two sides of a triangle are together greater than the third side.

4. The difference between any two sides of a triangle is less than the third side.

5. If three sides of one triangle are respectively equal to three sides of another triangle, then such triangles are congruent.

6. If a side and two adjacent angles of one triangle are respectively equal to a side and two adjacent angles of another triangle , then such triangles are congruent.

7. If two sides and the included angle of one triangle are respectively equal to two sides and the included angle of another triangle, then such triangles are congruent.

8. The sum of any three angles in any triangle is equal to 180°.

9. An exterior angle is equal to the sum of the non adjacent interior angles.

10. If two sides of a triangle are unequal then the greater side has the greater angle opposite to it.

11. If one angle of a triangle is greater than other then the side opposite to the greater angle is greater than the side opposite to the lesser angle.

12. **Basic proportionality theorem(Thales theorem):** In a triangle a line drawn parallel to one side , to intersect the othersides in distinct points divides the two sides in the same ratio.

13. **Converse of basic proportionality theorem:** If a line divides any two sides of a triangle in the same ratio , the line must be parallel to the third side.

14. The straight line joining the midpoints of any two sides of a triangle is parallel to the third and half of it.

15. The ratio of the areas of two similar triangles is equal to the ratio of the squares of any two corresponding sides.

16. The areas of two similar triangles are in the ratio of the squares of the corresponding altitudes.

17. The areas of two similar triangles are in the ratio of the squares of the corresponding altitudes.

18. The area of two similar triangles are in the ratio of the squares of the corresponding medians.

19. The areas of two similar triangles are in the ratio of the squares of the corresponding angle bisector segments.

20. If two sides and a median bisecting one of these sides of a triangle are respectively proportional to the two sides and the corresponding median of another triangle, then the triangles are similar.

21. If two triangles are equiangular, then the ratio of the corresponding sides is the same as the ratio of corresponding medians.

22. If two triangles are equiangular, then the ratio of the corresponding sides is the same as the ratio of corresponding angle bisector segments.

23. The internal bisector of an angle of a triangle divides the opposite side internally in the ratio of the sides containing angle.

24. The bisector of the exterior angle A of a triangle ABC intersects the side BC produced in D then $\frac{AB}{AC} = \frac{BD}{CD}$.

25. If a perpendicular is drawn from the vertex containing the right angle of a right triangle to the hypotenuse, then the triangles on each side of the perpendicular are similar to each other and to the original triangle and also the square of the perpendicular is equal to the product of the lengths of the two parts of the hypotenuse.

26. A) ***Pythagoras theorem***: In a right triangle, the square of the hypotenuse is equal to the sum of the squares on the other two sides.

    B) ***Converse of Pythogoras theorem***: If the square of the one side of a triangle is equal to the sum of the squares on the other two sides, then these two sides contain a right angle.

    C)In DABC, $\angle B < 90°$, if $AD \perp BC$; then $b^2 = a^2 + c^2 - 2ax = BD$.

    D)DABC in which AE is a median on BC and $AD \perp BC$.

    then $AB^2 + AC^2 = 2AE^2 + 2BE^2$. This is called Appolonius theorem.

27. If two arcs of a circle are congruent then the corresponding chords are equal.

28. If two chords of a circle are equal, then their corresponding arcs are congruent.
29. The perpendicular from the centre of a circle to a chord bisects the chord.
30. The line joining the centre of a circle to the midpoint of a chord is perpendicular to the chord.
31. Equal chords of a circle are equidistant from the centre.
32. Chords of a circle , which are equidistant from the centre are equal.
33. Any two angles in the same segment of a circle are equal.
34. An angle in a semicircle is a right angle.
35. The arc of a circle subtending a right angle at any point of the circle in alternate segment is a semi circle.
36. If a line segment joining two points subtends equal angles at two other points lying on the same side of the line containing the same segment , the four points are concyclic.
37. Equal chords of a circle subtend equal angles at the centre.
38. If the angles subtended by two chords of a circle at the centre are equal, the chords are equal.
39. The sum of either pair of opposite angles of a cyclic quadrilateral is 180°
40. A tangent to a circle is perpendicular to the radius through the point of contact.
41. A line drawn through end point of a radius and perpendicular to its tangent to the circle.
42. If two chords of a circle intersect inside or outside of the circle when produced , the rectangle formed by the two segments of one chord is equal in area to the rectangle formed by the two segments of the other.
43. Parallelograms and rectangles on the same base and between the same parallels are equal in area. Also their altitudes are equal.
44. A line drawn parallel to one side of a triangle divides the other two sides forming another triangle. Such a triangle is congruent to the given triangle.

45. If two chords of a circle intersect at a point inside the circle, then the rectangle contained by the segments of chord is equal to the rectangle contained by the segment of the other chord.

46. If PAB is a secant to a circle intersecting the circle at A and B and PT is a tangent segment then $PA.PB = PT^2$

47. If a chord is drawn through the point of contact of a tangent to the circle, then the angles which this chord makes with the given tangent are equal respectively to the angles formed in the corresponding alternate segments.

48. If a line is drawn through an end point of a chord of a circle so that the angle formed with the chord is equal to the angle -subtended by the chord in the alter nate segment then the line is tangent to the circle.

49. If two circle touch each other the point of contact lies on the line passing through the centers.

50. Diagonals of trapezium divide each other proportionally.

51. If diagonals of a quadrilateral divide each other proportionally then it is trapezium.

52. **Tolmeys theorem:**If a quadrilateral is inscribed in a circle, the sum of the products of the two pairs of opposite sides is equal to the diagonals.

53. **Cevas theorem:**The three cevians AD, BE and CF of triangle are concurrent if and only if $\frac{AF}{FB}.\frac{BD}{DC}.\frac{CE}{EA}=1$

54. **Stewarts Formulae**.The length of a cevian 'x' and the sides of a triangle are related by the following formulae $c^2n + b^2m = x^2a + mna$.

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19. If $A=\begin{bmatrix}1 & 2\\ 2 & 1\end{bmatrix}$, then adj A =

1) $\begin{bmatrix}-1 & 2\\ 2 & -1\end{bmatrix}$ 2) $\begin{bmatrix}1 & -2\\ -2 & 1\end{bmatrix}$ 3) $\begin{bmatrix}2 & 1\\ 1 & 1\end{bmatrix}$ 4) $\begin{bmatrix}1 & -2\\ -2 & -1\end{bmatrix}$

20. If $A=\begin{bmatrix}1 & 3\\ 2 & -2\end{bmatrix}$, then A^{-1} =

1) $-\frac{1}{8}\begin{bmatrix}-2 & -3\\ -2 & 1\end{bmatrix}$ 2) $-\frac{1}{8}\begin{bmatrix}3 & 1\\ -2 & 2\end{bmatrix}$ 3) $\frac{1}{8}\begin{bmatrix}2 & 3\\ 2 & -1\end{bmatrix}$ 4) $\frac{1}{8}\begin{bmatrix}1 & 3\\ 2 & 2\end{bmatrix}$

21. If B is a non singlular matrix and A is a square matrix, then $\det\left(B^{-1}AB\right)$ is equal to

1) $\det\left(A^{-1}\right)$ 2) $\det\left(B^{-1}\right)$ 3) $\det(A)$ 4) $\det(B)$

22. If $A=\begin{bmatrix}\alpha & 0\\ 1 & 1\end{bmatrix}$ and $B=\begin{bmatrix}1 & 0\\ 5 & 1\end{bmatrix}$, then value of α for which $A^2=B$, is

1) 1 2) -1 3) 4 4) no real values

23. If the system of equations x + ay = 0, az + y = 0 and ax + z = 0 has infinite solutions, then the value of a is

1) -1 2) 1 3) 0 4) no real values

24. If $A=\begin{bmatrix}\alpha & 2\\ 2 & \alpha\end{bmatrix}$ and $\left|A^3\right|$ = 125 then the value of α is

1) ± 1 2) ± 2 3) ± 3 4) ± 5

25. If A and B are two square matrices such that B = $-A^{-1}$ BA, then $(A+B)^2=$

1) 0 2) A^2+B^2 3) $A^2+2AB+B^2$ 4) A+ B

26. If A = [a_{ij}] , i , j = 1, 2 where a_{ij} is defined as $a_{ij} = i^2 + j^2$ then A =

1) $\begin{bmatrix}2 & 5\\ 5 & 8\end{bmatrix}$ 2) $\begin{bmatrix}2 & 5\\ 4 & 8\end{bmatrix}$ 3) $\begin{bmatrix}2 & 4\\ 5 & 8\end{bmatrix}$ 4) $\begin{bmatrix}2 & 5\\ 5 & 7\end{bmatrix}$

27. If $A^T + B = \begin{bmatrix}2 & 1\\ 3 & 4\\ 5 & 6\end{bmatrix}$, $A - B^T = \begin{bmatrix}2 & 3 & 3\\ 7 & 6 & 2\end{bmatrix}$, then A =

1) $\begin{bmatrix}2 & 3 & 4\\ 4 & 5 & 4\end{bmatrix}$ 2) $\begin{bmatrix}2 & 3 & 4\\ 4 & 5 & 6\end{bmatrix}$ 3) $\begin{bmatrix}2 & 5 & 4\\ 4 & 5 & 4\end{bmatrix}$ 4) $\begin{bmatrix}2 & 3 & 5\\ 4 & 5 & 5\end{bmatrix}$

28. If order of A , B , C are 3 x 4, 5 x 4, 5 x 8 then what is the order of (AB^T3)

1) 2 X 4 2)3 x 8 3) 3X 5 4) 3X 6

10. If $\sin(\theta+\alpha)=\cos(\theta+\alpha)$, then $\tan\theta$ in terms of α is

1) $\frac{1+\tan\alpha}{1-\tan\alpha}$ 2) $\frac{1-\tan\alpha}{1+\tan\alpha}$ 3) $\frac{\tan\alpha-1}{\tan\alpha+1}$ 4) $\frac{\tan\alpha+1}{\tan\alpha-1}$

11. The value of $\frac{1-\tan^2(45^\circ-\theta)}{1+\tan^2(45^\circ+\theta)}$ is equal to

1) $\sin\theta$ 2) $\cos 2\theta$ 3) $\sin 2\theta$ 4) $\cos\theta$

12. If $2\cos\theta = a+\frac{1}{a}$, then $a^2+\frac{1}{a^2}$ is equal to

1) $\cos 2\theta$ 2) $\sin 2\theta$ 3) $2\cos 2\theta$ 4) $2\sin 2\theta$

13. If $\text{Sin}x+\text{Sin}^2x+\text{Sin}^3x=1$ then $\text{Cos}^6x - 4\text{Cos}^4x+8\text{Cos}^2x =$

1) 3 2) 4 3) 2 4) 1

14. If A = $\sin 45^0 + \cos 45^0$ and B = $\sin 44^0 + \cos 44^0$ then,

1)A > B 2)A < B 3)A = B 4)none

15. The value of $\tan 9^0 - \tan 27^0 - \tan 63^0 + \tan 81^0$ is

1)2 2)3 3)4 4)1

16. $2(\sin^6 A + \cos^6 A) - 3(\sin^4 A + \cos^4 A)$ is,

1)2 2)0 3)-1 4)1

17. The value of the expression $\sqrt{3}\,\text{cosec}\,20^0 - \sec 20^0$ is equal to

1)2 2)$2\sin 20^0$ 3)4 4)$4\sin 20^0$

18. If cot (a + b) = 0, then sin (a + 2b) =

1)- sin a 2)sin b 3)cos a 4)cos b

19. Tan1° Tan2° Tan3°.....Tan89°=

1)1 2)2 3)0 4)21

20. If $2\cos\theta = x+\frac{1}{x}$ then $\cos\theta =$

1)1 2)0 3)0.5 4)0.4

21. If $\sin x+\sin y+\sin z=3$ then find value of $\cos x+\cos y+\cos z$

1)0 2)1 3)2 4)3

22. If $\sin(A-B)=\frac{3}{5}$ and $\sin(A+B)=\frac{4}{5}$ then find the value of $\sin 2A$.

1)0 2)1 3)2 4)3

23. If x = cos 1, y = $\cos 1^0$ then

10. If all the terms of an algebraic expression are of the same degree then such an expression is called a _____ expression.

1) symmetric 2) similar

3) congruent 4) homogenous expression

11. $ax^3 + bx^2y + cxy^2 + dy^3$ is a _____ degree homogenous expression

1) 1st 2) 2nd 3) 4th 4) 3rd

12. An algebraic expression f(x, y) in two variables x, y is called a___ expression, if f(x, y) = f(y, x)

1) homogenous 2) simple 3) symmetric 4) similar

13. Identify the symmetric expression among the following

1) $2x + 5 + 2y$ 2) $2x^2 + 5 + 3y^2$ 3) $2x^3 + 5 + 3y^2$ 4) $x + y^2 + 5$

14. $2(x^2 + y^2) + 3(x + y) + 7$ is

1) homogeneous 2) symmetric

3) not symmetric 4) homogeneous symmetric

15. Which of the following statement is false?

1) $ax + by + c$ is a non homogeneous

2) $x^3 - 3x^2y + 3xy^2 + y^3$ is homogeneous

3) $a^2(b + 3) + b^2(c + 1) + c^2(a + 2)$ is symmetric

4) $x^3 + 3x^3y + xy^2 + y^3$ is symmetric

16. If $ax^3 + bx^2y + cxy^2 + dy^3$ is a symmetric expression, then

1) a = b and c = d 2) a = c and b = d 3) a = d and b = c 4) $a \neq d$ and $c \neq d$

17. If ax + by is a homogeneous expression of first degree in x and y. When x = 1, y = 2 the value of expression is 9 and when x = 2, y = 3 the value of expression is 16. Then the expression is

1) $2x + 5y$ 2) $5x + 2y$ 3) $5x + 4y$ 4) $5x + 3y$

18. Which of the algbraic expressions $A = x^3 + 4x^2y + 5xy^2 + y^3$,
$B = x^3 + y^3 + x^2 + y^2 + x + y$,

$C = a^2(b + 3) + b^2(c + 1) + c^2(a^2 + b^3)$,

$D = 2x^2 + 3y^3 + x^3 + y^2 + 1$ is symmetric ?

1) A 2) B 3) C 4) D

19. One of the factor of $\sum_{a,b,c} ab\left(a^2 - b^2\right)$ is

1) (b + 3) 2) (a + 3) 3) a + b + c 4) none

20. One of the factor of $\sum_{a,b,c} a\left(b^4 - c^4\right)$ is

1) (c + 1) 2) (c + 2) 3) (b + 1) 4) (c – 1)

12. If divisor, quotient and remainder are $2x^2 - 6x + 7$, $3x - 2$ and $-6x + 5$, then the dividend is

1) $6x^3+22x^2+27x+9$ 2) $6x^3+22x^2-27x+9$
3) $6x^3-22x^2+27x-9$ 4) $6x^3-22x^2+27x+9$

13. If $x^2 + 2x - 63$ is exactly divided by a divisor and quotient is $x - 7$ then the divisor is

1) $x - 9$ 2) $x - 7$ 3) $x + 9$ 4) $x + 3$

14. If $v+\frac{1}{v}=5$, then $v^4+\frac{1}{v^4}$ =

1) 625 2) 729 3) 527 4) 841

15. If $x+\frac{1}{x}=10$, then $x-\frac{1}{x}$ =

1) $\pm4\sqrt{6}$ 2) $\pm2\sqrt{6}$ 3) $\pm3\sqrt{6}$ 4) $\pm5\sqrt{6}$

16. If $k^2 - 3k + 1 = 0$, then $k+\frac{1}{k}$ =

1) 3 2) 9 3) 27 4) 81

17. If $a^2-5a+1=0$, then $a^2+\frac{1}{a^2}$ =

1) 25 2) 23 3) 27 4) 22

18. If $x-\frac{1}{x}=5$, then $x^3-\frac{1}{x^3}$ =

1) 100 2) 125 3) 140 4) 145

19. If $x + y = 8$, $xy = 12$, then $x^4+y^4=$

1) 1012 2) 1112 3) 1212 4) 1312

20. If $z-\frac{1}{z}=4$, then $z^2+\frac{1}{z^2}=$

1) 18 2) 16 3) 14 4) 8

21. The product of $(x + 1)(x + 2)(x - 3)$ is

1) $x^2 - 6$ 2) $x^3 - 7x - 6$ 3) $x^3 + 6x^2 - 7x - 6$ 4) $x^3 - 6x^2 - 7x - 6$

22. If $a + b + c = 0$ then $a^3 + b^3 + c^3 =$

1) 3 2) 3abc 3) abc 4) 9abc

23. Which of the following is not a polynomial?

1) $x^2 + 3x + 5$ 2) $x^2 + 3x + \frac{5}{x}$ 3) $0 - x^4 + x^2 - x$ 4) 5

24. The value of b in the equation $(x + 6)(x + 2) = x^2 + 2x - 24$ is

1) 2 2) 1. 3) – 2 4) – 4

25. If $(x + 1)^2 = x$, then $11x^3 + 8x^2 + 8x - 2 =$

1) 1 2) 2 3) 3 4) 4

4. The total number of distinct prime factors in the expression

$4^{11} \times 7^5 \times 11^2$ is

1) 18 2) 29 3) 20 4) 3

5. A number when divided by 114 leave the remainder 21. If the same number is divided by 19, the remainder will be

1) 1 2) 2 3) 7 4) 21

6. In doing a division of a question with zero remainder, a candidate took 12 as divisor instead of 21. The quotient obtained by him was 35. The correct quotient is

1) 0 2) 12 3) 13 4) 20

7. The least number which divided by 5, 6, 7, 8 leaves a remainder 3, but when divided by 9 leaves no remainder is

1) 1677 2)1683 3) 2523 4) 3363

8. Find the largest number which divides 62, 132 and 237 to leave the same remainder in each case.

1) 15 2) 27 3) 35 4) none of these

9. When an even perfect square divided by 4 will leave which type of number as a quotient

1) even number 2) odd number

3) perfect square 4) prime number

10. The remainder when $10^1 + 10^2 + 10^3 + 10^4 + \ldots\ldots\ldots + 10^{99}$ is divided by 6

1) 1 2) 2 3) 0 4) 3

11. Which of the following numbers are of the form $4n(n + 1) + 1$

1) 4489 2) 444889 3) 44448889 4) all of these

12. The remainder when an odd perfect square is divided by 8

1) 0 2) 1 3) 2 4) 3

13. The G.C.D of the number groups

$(3^{2010} - 1)$ and $(3^{2010} + 1)$; $(5^{2010} - 1)$ and $(5^{2010} + 1)$

$(7^{2010} - 1)$ and $(7^{2010} + 1)$; $(9^{2010} - 1)$ and $(9^{2010} + 1)$; and so on till

$(2009^{2010} - 1)$ and $(2009^{2010} + 1)$ is

1) different in each case 2) same in all those different cases

10. If $a = \log_{12}18$ & $b = \log_{24}54$ then find the value of $ab + 5(a - 2)$.

1)1 2)2 3)3 4)4

11. If $x = 1 + \log_a bc$, $y = 1 + \log_b ca$, $z = 1 + \log_c ab$ then $xy + yz + zx - xyz =$

1)0 2)1 3)2 4)3

12. If $p = \log_a bc$, $q = \log_b ca$, $r = \log_c ab$, then $pqr - p - q - r =$

1)1 2)2 3)3 4)5

13. If $(5.3)^x = (0.053)^y = 1000$ then the value of $\frac{1}{x} - \frac{1}{y} =$

1) $\frac{8}{3}$ 2) $\frac{4}{3}$ 3) $\frac{2}{3}$ 4) $\frac{1}{3}$

14. If $\frac{\log x}{y-z} = \frac{\log y}{z-x} = \frac{\log z}{x-y}$ then value of $x^x y^y z^z$

1)1 2)2 3)3 4)5

15. $\frac{1}{\log_2 a} + \frac{1}{\log_3 a} + \frac{1}{\log_4 a} + \ldots\ldots\ldots\ldots + \frac{1}{\log_k a} =$

1) $\log_{k!} a$ 2) $\log_a k!$ 3) $\log_a k$ 4)0

16. $\frac{1}{\log_{\frac{a}{b}} x} + \frac{1}{\log_{\frac{b}{c}} x} + \frac{1}{\log_{\frac{c}{a}} x} =$

1) 1 2) abc 3) Not defined 4) 0

17. If $\log\left(\frac{x+y}{5}\right) = \frac{1}{2}(\log x + \log y)$ then $\frac{x}{y} + \frac{y}{x} =$

1)21 2)23 3)25 4)15

18. If $\log_k x \cdot \log_5 k = \log_x 5,\ k \neq 1, k > 0$, then x =

1)k 2)5 3)10 4) $\frac{1}{5}$

19. The least value of the expression $2\log_{10} x - \log_x(0.01)$, for x >1, is:

1) 10 2)2 3) -0.01 4) 4

20. The number of solutions of $\log_2 (x + 5) = 6 - x$ is

1) 2 2) 0 3) 3 4) 1

SETS

1. **Set :** A set is collection of well defined objects.(There is no precise mathematical definition).Sets are denoted by A , B , C , X ,Y .....

2. **Representation of sets:**

1) **Roster method(tabular form):**A set may be specified by listing all of its elements seperating them by commas and enclosing them within brackets.

Ex :The set of all solutions of $x^2 - 4 = 0$ is specified by { -2, 2 }

2) **Set-Builder method:**A set can be described by stating some property of the element as $X = \{ x \mid P(x) \}$

Ex: The set of all solutions of $x^2 - 4 = 0$ is specified by

$X = \{ x \mid x \in Z \text{ and } x^2 - 4 = 0\}$ or as $X = \{ x \in Z \mid x^2 - 4 = 0\}$

3. **Finite Set:** A set is called a finite set if it consists of a finite number of elements.

4. **Infinite Set:** A set which is not a finite set is called an infinite set.

5. **Cardinal Number of Set**: The number of elements in a finite set is called the Cardinal number of the set. The cardinal number of a set A is denoted by **n(A).**

6. **Singleton Set:** A set consisting of only one element is called a singleton set.

7. **Empty Set:** A set having no element in it is called **EMPTY or VOID or** NULL SET. The empty set is denoted by the symbol f or { }.

8. **Sub Set:** A set A is called a subset of the set B, if every element of A is also an element of B.It is denoted by '$\subseteq$', read as 'is a subset of '

9. 1.Empty set is a subset of every set.

2. Every set is a subset to itself.

10. **Super Set:** If $A \subseteq B$ then B is called super set of A. Symbolically 'is a superset denoted by '$\supseteq$' .

11. **Equal Sets**: Two sets A and B are said to be equal or identical if (i) A is a subset of B, and (ii) B is a subset of A.

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X FOUNDATION MATHEMATICS

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